Code:

```
interface ParkingCharge {
    double calculateCharge();
    void displayParkingDetails();
}
class VehicleParking {
    String vehicleNumber;
    String vehicleType;
    int parkingHours;
    public VehicleParking(String vehicleNumber, String
vehicleType, int parkingHours) {
        this.vehicleNumber = vehicleNumber;
        this.vehicleType = vehicleType;
        this.parkingHours = parkingHours;
    }
}
class TwoWheeler extends VehicleParking implements
ParkingCharge {
    public TwoWheeler(String vehicleNumber, int parkingHours)
{
        super(vehicleNumber, "Two Wheeler", parkingHours);
    }
   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 20.0;
        } else {
            return (2 * 20.0) + ((parkingHours - 2) * 10.0);
        }
    }
   public void displayParkingDetails() {
        System.out.println("Vehicle Number: " +
```

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vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours +
" hrs");
        System.out.println("Total Parking Charge: Rs. " +
calculateCharge());
        System.out.println(" ");
    }
}
class FourWheeler extends VehicleParking implements
ParkingCharge {
    public FourWheeler(String vehicleNumber, int parkingHours)
{
        super(vehicleNumber, "Four Wheeler", parkingHours);
    }
   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 50.0;
        } else {
            return (2 * 50.0) + ((parkingHours - 2) * 30.0);
        }
    }
    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " +
vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " + calculateCharge());
        System.out.println(" ");
    }
}
```

```
public class ParkingManagementSystem {
    public static void main(String[] args) {
        ParkingCharge parkingRef;
        parkingRef = new TwoWheeler("KA-01-AB-1234", 5);
        parkingRef.displayParkingDetails();
        parkingRef = new FourWheeler("KA-05-CD-5678", 4);
        parkingRef.displayParkingDetails();
        System.out.println("Coded by: Kasham Adhikari");
        System.out.println("Section: C");
        System.out.println("USN: 1BF25CS122");
    }
}
```

Output:

```
----jGRASP exec: java ParkingManagementSystem
Vehicle Number: KA-01-AB-1234
Vehicle Type: Two Wheeler
Parking Hours: 5 hrs
Total Parking Charge: Rs. 70.0

Vehicle Number: KA-05-CD-5678
Vehicle Type: Four Wheeler
Parking Hours: 4 hrs
Total Parking Charge: Rs. 160.0

Code by: Kasham Adhikari
Section: C
USN: 1BF25CS122

----jGRASP: Operation complete.
```